

Practice Questions for Gordan's Theorem

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1 Questions

1. Gordan's Theorem establishes a fundamental relationship between the solvability of a system of linear inequalities and the existence of a solution to a related system. Which of the following best describes the core idea of Gordan's Theorem?
 - a. A system of linear inequalities $Ax \leq 0$ has a solution if and only if the system $A^T y = 0, y \geq 0$ has no solution.
 - b. A system of linear inequalities $Ax < 0$ has a solution if and only if the system $A^T y = 0, y \geq 0$ has a solution.
 - c. A system of linear inequalities $Ax \leq 0$ has a solution if and only if the system $A^T y = 0, y > 0$ has a solution.
 - d. A system of linear inequalities $Ax < 0$ has a solution if and only if the system $A^T y = 0, y \geq 0$ has no solution.
 - e. A system of linear inequalities $Ax \leq 0$ has no solution if and only if the system $A^T y = 0, y > 0$ has a solution.
2. Gordan's Theorem is a fundamental result in linear algebra with significant implications for linear programming. What is the key implication of Gordan's Theorem regarding the feasibility of a linear program?
 - a. It guarantees the existence of an optimal solution for every linear program.
 - b. It provides a condition for determining whether a system of linear inequalities defining a linear program's feasible region is consistent (has a solution).
 - c. It establishes a direct relationship between the primal and dual objective function values.
 - d. It determines the uniqueness of the optimal solution to a linear program.
 - e. It provides a method for finding the optimal solution to a linear program.
3. Consider the system of linear inequalities $Ax \leq 0$. How does Gordan's Theorem relate the existence of a non-trivial solution ($x \neq 0$) to this system and the existence of a solution to another related system?
 - a. If $Ax \leq 0$ has a non-trivial solution, then $A^T y = 0, y \geq 0$ must also have a non-trivial solution.
 - b. If $Ax \leq 0$ has no non-trivial solution, then $A^T y = 0, y > 0$ must have a solution.
 - c. If $Ax \leq 0$ has a non-trivial solution, then $A^T y = 0, y > 0$ must have no solution.
 - d. If $Ax \leq 0$ has no non-trivial solution, then $A^T y = 0, y \geq 0$ must have a solution.

- e. The two systems are independent; the existence of a solution to one does not imply anything about the other.
4. Gordan's Theorem is often used to prove other fundamental theorems in linear programming. Which of the following theorems is most directly related to and often proven using Gordan's Theorem?
- a. Strong Duality Theorem
 - b. Weak Duality Theorem
 - c. Farkas' Lemma
 - d. Carathéodory's Theorem
 - e. Separation Theorem
5. Gordan's Theorem establishes a fundamental relationship between two systems of linear inequalities. Explain the core idea of Gordan's Theorem and its implications for the feasibility of a linear program.
- a. Gordan's Theorem states that if a system of homogeneous linear inequalities $Ax \leq 0$ has a non-trivial solution, then the system $A^T y = 0, y > 0$ has no solution, and vice versa. This implies that a linear program is either feasible or its dual is feasible, but not both simultaneously.
 - b. Gordan's Theorem states that if a system of linear inequalities $Ax \leq b$ has a solution, then the system $A^T y = 0, b^T y > 0$ has no solution, and vice versa. This implies that a linear program is feasible if and only if its dual is infeasible.
 - c. Gordan's Theorem states that if a system of homogeneous linear equations $Ax = 0$ has a non-trivial solution, then the system $A^T y = 0$ has only the trivial solution, and vice versa. This has no direct implication for the feasibility of a linear program.
 - d. Gordan's Theorem states that if a system of linear inequalities $Ax \leq b$ has a solution, then the system $A^T y = c$ has a solution, and vice versa. This implies that a linear program is feasible if and only if its dual is feasible.
 - e. Gordan's Theorem states that if a system of linear equations $Ax = b$ has a solution, then the system $A^T y = 0$ has only the trivial solution, and vice versa. This has no direct implication for the feasibility of a linear program.

2 Answers

1. Gordan's Theorem establishes a fundamental relationship between the solvability of a system of linear inequalities and the existence of a solution to a related system. Which of the following best describes the core idea of Gordan's Theorem?
 - a. Gordan's Theorem states that exactly one of the following systems has a solution: $Ax \leq 0$ with $x \neq 0$ or $A^T y = 0$ with $y > 0$. Option A correctly reflects this by stating that if one system has a solution, the other does not. The strict inequality $y > 0$ is relaxed to $y \geq 0$ and $x \neq 0$ to $x \leq 0$ in some formulations.
 - b. This option incorrectly reverses the relationship between the two systems. If $Ax < 0$ has a solution, it does not imply that $A^T y = 0, y \geq 0$ necessarily has a solution.
 - c. This option introduces a strict inequality ($y > 0$) in the second system, which is not the standard formulation of Gordan's Theorem.
 - d. This option incorrectly states that if $Ax < 0$ has a solution, then $A^T y = 0, y \geq 0$ has no solution. This is not true in general.
 - e. This option is close but not quite correct. The correct statement involves $y > 0$ in the second system, not $y \geq 0$.
2. Gordan's Theorem is a fundamental result in linear algebra with significant implications for linear programming. What is the key implication of Gordan's Theorem regarding the feasibility of a linear program?
 - a. Gordan's Theorem does not guarantee the existence of an optimal solution; a linear program may be infeasible or unbounded.
 - b. Gordan's Theorem directly addresses the consistency of a system of linear inequalities. In linear programming, the feasible region is defined by such a system, so the theorem helps determine if the feasible region is non-empty.
 - c. Gordan's Theorem does not directly relate to the primal and dual objective function values; that's the domain of strong and weak duality theorems.
 - d. Gordan's Theorem doesn't address the uniqueness of the optimal solution; multiple optimal solutions are possible in linear programming.
 - e. Gordan's Theorem is a theoretical result; it doesn't provide a method for finding the solution.
3. Consider the system of linear inequalities $Ax \leq 0$. How does Gordan's Theorem relate the existence of a non-trivial solution ($x \neq 0$) to this system and the existence of a solution to another related system?
 - a. This is incorrect; the existence of a solution to one system does not guarantee a solution to the other in this way.
 - b. This is incorrect; the condition is $y > 0$, not $y \geq 0$.
 - c. Gordan's Theorem states that exactly one of the systems has a solution: either $Ax \leq 0$ with $x \neq 0$ or $A^T y = 0$ with $y > 0$. Therefore, if the first system has a non-trivial solution, the second must have no solution.
 - d. This is incorrect; the condition is $y > 0$, not $y \geq 0$.
 - e. This is incorrect; Gordan's Theorem explicitly establishes a relationship between the two systems.

4. Gordan's Theorem is often used to prove other fundamental theorems in linear programming. Which of the following theorems is most directly related to and often proven using Gordan's Theorem?
- a. While related to duality, the Strong Duality Theorem is not directly proven using Gordan's Theorem.
 - b. The Weak Duality Theorem is a simpler result and doesn't typically rely on Gordan's Theorem for its proof.
 - c. Farkas' Lemma is closely related to Gordan's Theorem and is often proven using it. Both theorems deal with the solvability of systems of linear inequalities and their implications.
 - d. Carathéodory's Theorem deals with convex hulls and is not directly related to Gordan's Theorem.
 - e. The Separation Theorem deals with separating hyperplanes and is not directly proven using Gordan's Theorem.
5. Gordan's Theorem establishes a fundamental relationship between two systems of linear inequalities. Explain the core idea of Gordan's Theorem and its implications for the feasibility of a linear program.
- a. Gordan's Theorem directly addresses the feasibility of a system of homogeneous inequalities. If one system has a non-trivial solution, the other has no solution, highlighting a fundamental duality relationship crucial in linear programming. The statement about a linear program and its dual being either feasible or infeasible is incorrect; both can be infeasible.
 - b. This option incorrectly mixes homogeneous and inhomogeneous systems. Gordan's Theorem deals with homogeneous systems ($Ax \leq 0$). Also, a linear program can be infeasible even if its dual is infeasible.
 - c. This option describes a different theorem related to the existence of solutions to homogeneous systems, not Gordan's Theorem.
 - d. This option incorrectly states that the feasibility of a linear program implies the feasibility of its dual, which is not always true. Both can be infeasible.
 - e. This option describes a different theorem related to the existence of solutions to linear systems, not Gordan's Theorem.